

Binomial Distribution

ANSWER KEY



Conditions for a binomial distribution

- 1 State the four conditions required for a random variable to follow a binomial distribution.
(1) A fixed number of trials n . (2) Each trial has only two outcomes (success or failure). (3) The probability of success p is constant for every trial. (4) The trials are independent of each other.
- 2 A survey asks 50 people whether they support a new policy, and X is the number who say yes, assuming each person's response is independent with the same probability of support. Explain why X follows a binomial distribution.
 X follows a binomial distribution because there is a fixed number of trials ($n = 50$), each trial (person) has two outcomes (yes or no), the probability of a 'yes' response is constant across all people, and the responses are independent of each other — satisfying all four binomial conditions.
- 3 A researcher rolls a die 15 times and counts the number of times a 3 appears. Explain why this count follows a binomial distribution.
This count follows a binomial distribution because there is a fixed number of trials ($n = 15$), each roll has two outcomes (rolling a 3 or not), the probability of rolling a 3 is constant ($\frac{1}{6}$) on every roll, and the rolls are independent of each other — satisfying all four binomial conditions.

Calculating binomial probabilities

- 4 A fair coin is tossed 8 times. Find the probability of getting exactly 5 heads, using
 $P(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}$.
 $X \sim B(8, 0.5)$. $P(X = 5) = \binom{8}{5} (0.5)^5 (0.5)^3 = 56(0.03125)(0.125) = 56(0.00390625) \approx 0.2188$.
- 5 A factory produces items with a defect rate of 10%. In a sample of 6 items, find the probability that exactly 2 are defective.
 $X \sim B(6, 0.1)$. $P(X = 2) = \binom{6}{2} (0.1)^2 (0.9)^4 = 15(0.01)(0.6561) \approx 0.0984$.
- 6 A fair coin is tossed 6 times. Find the probability of getting exactly 4 heads.
 $X \sim B(6, 0.5)$. $P(X = 4) = \binom{6}{4} (0.5)^4 (0.5)^2 = 15(0.0625)(0.25) = 0.234375$.
- 7 A factory produces items with a defect rate of 20%. In a sample of 5 items, find the probability that exactly 1 is defective.
 $X \sim B(5, 0.2)$. $P(X = 1) = \binom{5}{1} (0.2)^1 (0.8)^4 = 5(0.2)(0.4096) = 0.4096$.

Cumulative binomial probabilities

- 8 A biased coin has $P(\text{heads}) = 0.3$. It is tossed 5 times. Find $P(X \leq 1)$, where X is the number of heads.
 $P(X = 0) = \binom{5}{0}(0.3)^0(0.7)^5 = 0.16807$. $P(X = 1) = \binom{5}{1}(0.3)^1(0.7)^4 = 5(0.3)(0.2401) = 0.36015$.
 $P(X \leq 1) = 0.16807 + 0.36015 \approx 0.5282$.
- 9 Using the same distribution $X \sim B(5, 0.3)$, find $P(X \geq 2)$.
 $P(X \geq 2) = 1 - P(X \leq 1) = 1 - 0.5282 \approx 0.4718$.
- 10 A biased coin has $P(\text{heads}) = 0.2$. It is tossed 4 times. Find $P(X \leq 1)$, where X is the number of heads.
 $P(X = 0) = \binom{4}{0}(0.2)^0(0.8)^4 = 0.4096$. $P(X = 1) = \binom{4}{1}(0.2)^1(0.8)^3 = 4(0.2)(0.512) = 0.4096$.
 $P(X \leq 1) = 0.4096 + 0.4096 = 0.8192$.

Mean and variance of a binomial distribution

- 11 A random variable $X \sim B(40, 0.25)$. Find the mean and variance, using $E(X) = np$ and $\text{Var}(X) = np(1 - p)$.
 $E(X) = np = 40(0.25) = 10$. $\text{Var}(X) = np(1 - p) = 40(0.25)(0.75) = 7.5$.
- 12 A random variable $X \sim B(60, 0.2)$. Find the mean and variance.
 $E(X) = np = 60(0.2) = 12$. $\text{Var}(X) = np(1 - p) = 60(0.2)(0.8) = 9.6$.
- 13 A random variable $X \sim B(80, 0.1)$. Find the mean and variance.
 $E(X) = np = 80(0.1) = 8$. $\text{Var}(X) = np(1 - p) = 80(0.1)(0.9) = 7.2$.
- 14 A random variable $X \sim B(50, 0.6)$. Find the mean and variance.
 $E(X) = np = 50(0.6) = 30$. $\text{Var}(X) = np(1 - p) = 50(0.6)(0.4) = 12$.

Finding an unknown parameter

- 15 A random variable $X \sim B(n, 0.4)$ has mean $E(X) = 20$. Find n .
 $E(X) = np$: $20 = n(0.4)$, so $n = 50$.
- 16 A random variable $X \sim B(25, p)$ has mean $E(X) = 10$. Find p .
 $E(X) = np$: $10 = 25p$, so $p = 0.4$.
- 17 A random variable $X \sim B(n, 0.25)$ has variance $\text{Var}(X) = 9$. Find n .
 $\text{Var}(X) = np(1 - p)$: $9 = n(0.25)(0.75) = 0.1875n$, so $n = \frac{9}{0.1875} = 48$.
- 18 A random variable $X \sim B(n, 0.5)$ has variance $\text{Var}(X) = 12.5$. Find n .
 $\text{Var}(X) = np(1 - p)$: $12.5 = n(0.5)(0.5) = 0.25n$, so $n = \frac{12.5}{0.25} = 50$.

Using the binomial distribution to model a real scenario

- 19** A multiple-choice quiz has 10 questions, each with 4 options, only one of which is correct. A student guesses randomly on every question. Find the probability the student gets at least 8 questions correct.
 $X \sim B(10, 0.25)$. $P(X = 8) = \binom{10}{8}(0.25)^8(0.75)^2 \approx 45(0.0000153)(0.5625) \approx 0.000387$.
 $P(X = 9) = \binom{10}{9}(0.25)^9(0.75)^1 \approx 10(0.00000381)(0.75) \approx 0.0000286$.
 $P(X = 10) = (0.25)^{10} \approx 0.00000095$.
 $P(X \geq 8) \approx 0.000387 + 0.0000286 + 0.00000095 \approx 0.000417$.
- 20** A basketball player has a 70% free-throw success rate. Find the probability that the player makes exactly 3 out of 5 free throws.
 $X \sim B(5, 0.7)$. $P(X = 3) = \binom{5}{3}(0.7)^3(0.3)^2 = 10(0.343)(0.09) = 0.3087$.
- 21** An archer hits the target with probability 0.8 on each shot, independently. Find the probability that the archer hits the target exactly 4 times out of 5 shots.
 $X \sim B(5, 0.8)$. $P(X = 4) = \binom{5}{4}(0.8)^4(0.2)^1 = 5(0.4096)(0.2) = 0.4096$.

Distinguishing binomial situations from non-binomial situations

- 22** Explain why drawing 5 cards from a standard deck WITHOUT replacement and counting the number of aces drawn is NOT a binomial distribution, while drawing 5 cards WITH replacement would be.
 In a binomial distribution, the probability of success must remain constant across all trials, which requires independence between trials. When drawing without replacement, the probability of drawing an ace changes after each draw (since the deck composition changes), so the trials are not independent and p is not constant. Drawing with replacement restores this constant probability and independence, satisfying the binomial conditions.
- 23** Explain why the number of sixes rolled in 10 rolls of a fair die IS a binomial distribution, while the number of rolls needed to get the first six is NOT.
 The number of sixes in 10 rolls is binomial because there is a fixed number of trials (10), each roll has two outcomes (six or not six), the probability of a six is constant ($\frac{1}{6}$) each time, and the rolls are independent. The number of rolls needed to get the first six does not have a fixed number of trials — the trial count itself is the random variable — so it does not satisfy the binomial conditions (it follows a geometric distribution instead).
- 24** Explain why the number of heads obtained by flipping a fair coin exactly 20 times IS a binomial distribution, while the number of flips needed to get 3 heads is NOT.
 The number of heads in 20 flips is binomial because there is a fixed number of trials (20), each flip has two outcomes (heads or tails), the probability of heads is constant (0.5) on every flip, and the flips are independent. The number of flips needed to get 3 heads does not have a fixed number of trials — the trial count itself is variable — so it does not satisfy the binomial conditions (it follows a negative binomial distribution instead).

A well-designed word problem: quality control sampling

- 25** This question has two parts. A machine produces bolts, of which 5% are defective. A quality inspector randomly samples 20 bolts each hour. (a) Find the expected number of defective bolts in a sample of 20. (b) The inspector rejects the whole batch if 3 or more defective bolts are found in the sample. Find the probability that the batch is rejected, given $P(X = 0) \approx 0.3585$, $P(X = 1) \approx 0.3774$, and $P(X = 2) \approx 0.1887$ for $X \sim B(20, 0.05)$.
- (a) $E(X) = np = 20(0.05) = 1$ defective bolt. (b)
- $P(X \geq 3) = 1 - P(X \leq 2) = 1 - [0.3585 + 0.3774 + 0.1887] = 1 - 0.9246 = 0.0754$, so there is approximately a 7.5% chance the batch is rejected.